DMZ Segmentation and Firewall Enforcement Implementation

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Overview

This project demonstrates practical implementation of network segmentation and traffic isolation using both host-based and network-based firewall controls in a Cisco Packet Tracer environment.

The objective was to isolate an internal FTP server within a segmented zone while enforcing least privilege access controls.

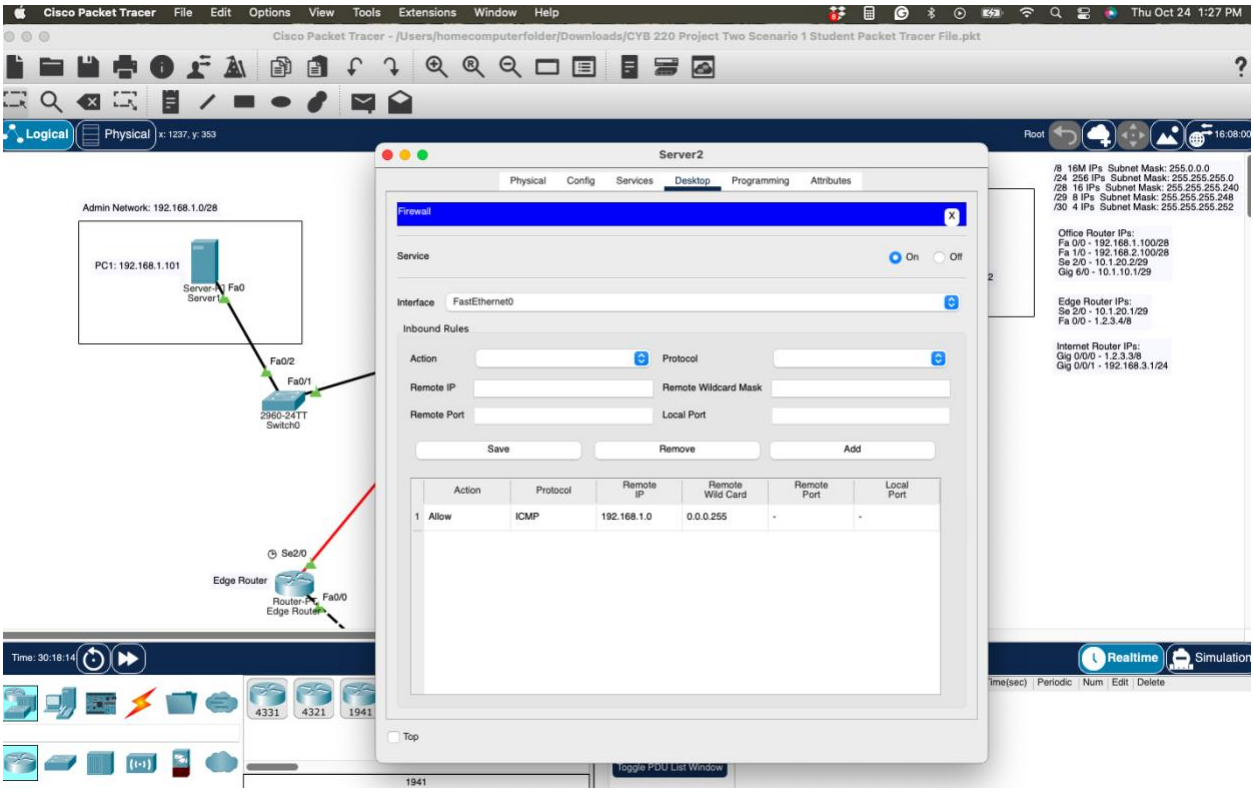
Environment Configuration

Host-Based Firewall Configuration

A host-based firewall policy was configured to enforce inbound traffic restrictions at the server level. Only authorized protocols and administrative traffic were permitted, while all unauthorized inbound traffic was denied.

This reduces the attack surface at the endpoint level and provides an additional layer of defense beyond perimeter controls.

(Screenshot: Host-Based Firewall Configuration)



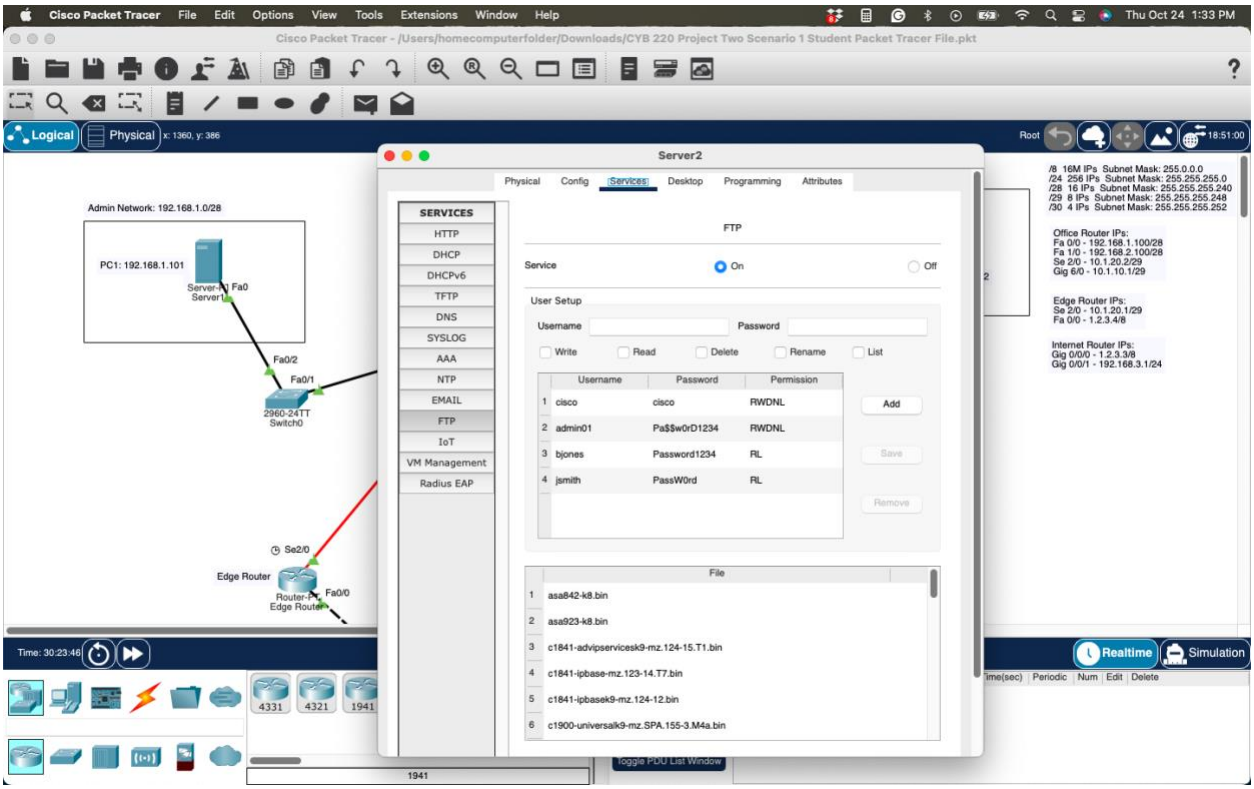
FTP Server Access Control

The FTP server was configured to allow access exclusively to the administrative subnet.

This configuration enforces the principle of least privilege by restricting file transfer access to only users with operational need.

All other network segments were prevented from accessing the FTP service.

(Screenshot: FTP Server Configuration)



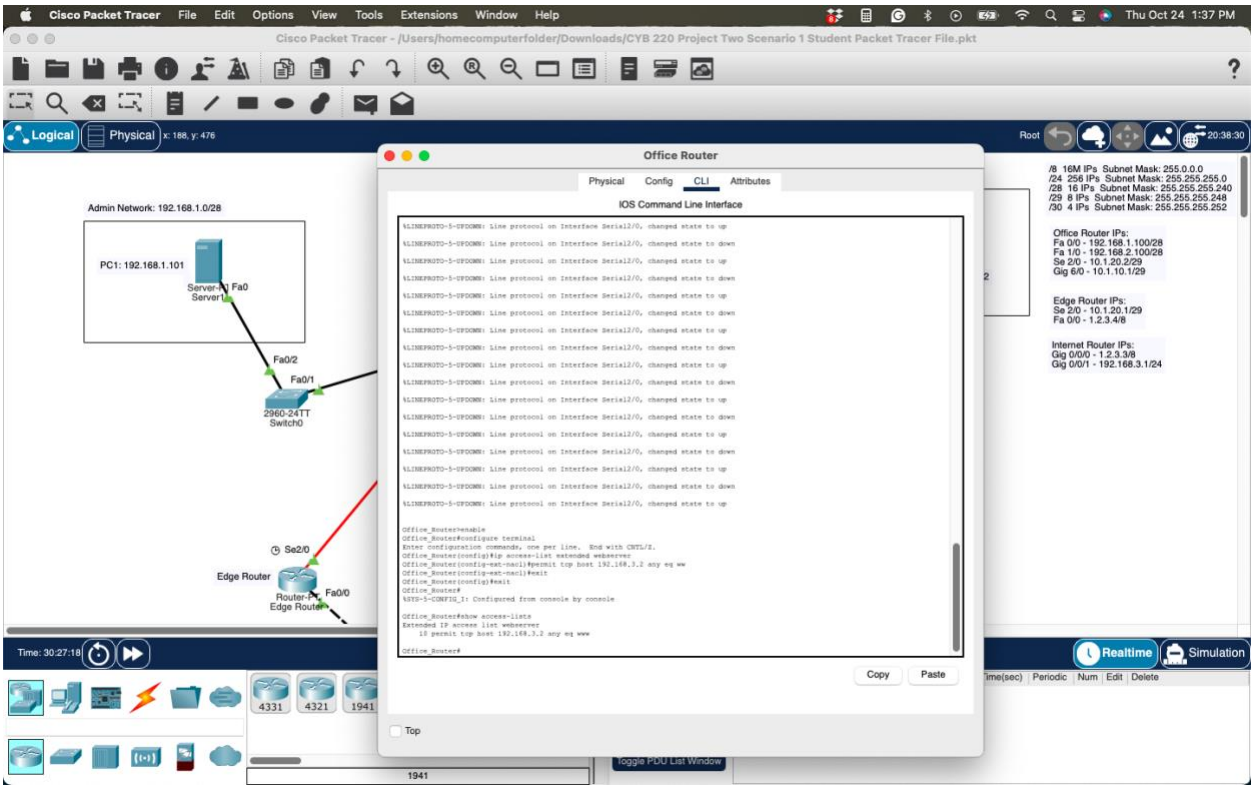
Network-Based Firewall (Extended ACL)

An extended Access Control List (ACL) was implemented on the edge router to:

- Permit FTP access from the admin subnet to the FTP server
- Redirect non-admin traffic to the public web server
- Deny unauthorized access attempts to the segmented zone

This ACL enforces network-layer isolation and ensures traffic is filtered before reaching protected assets.

(Screenshot: Extended ACL Configuration)



Security Outcomes

• Achieved logical and physical network segmentation
• Enforced least privilege access to sensitive services
• Implemented layered defense (host-based + network-based filtering)
• Reduced risk exposure through traffic validation

Tools Used

- Cisco Packet Tracer
- Extended Access Control Lists (ACLs)
- Host-based firewall policies
- Subnet segmentation